

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI -590 014



**Flipped class room activity report on
Design of Automatic Street lamp Using Arduino Uno R3 in
TinkerCAD simulation tool**

**DEPARTMENT
OF
ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted by

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S.J.M VIDYAPEETHA ®

S.J.M INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE,
New Delhi and Approved by Government of Karnataka)

NAAC Accredited with 'B⁺⁺' Grade

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1. Introduction

With the rapid growth of urbanization and increasing demand for electrical energy, efficient utilization of power has become a major concern. Street lighting is one of the essential public utilities that consumes a significant amount of electricity, especially in urban and rural areas. Conventional street lighting systems operate manually or based on fixed timers, which often results in unnecessary power wastage during daylight hours or low-traffic periods. To overcome these limitations, **automatic street light systems** have been introduced as a smart and energy-efficient solution.

Automatic street lighting is an energy-efficient system designed to control street lights automatically based on surrounding light conditions. Traditional street lights operate manually or remain ON throughout the night, leading to unnecessary power consumption. The automatic street light system uses a **Light Dependent Resistor (LDR)** and a microcontroller (Arduino) to detect ambient light intensity and switch the street lights ON during darkness and OFF during daylight. This project is designed and simulated using the TinkerCAD virtual designing tool, which helps in understanding real-time circuit behaviour without physical hardware.

2. Objectives

- To design an automatic street light control system using Arduino UNO R3.
- To understand the working of LDR sensors and analog inputs.
- To implement automatic ON/OFF control of lights based on light intensity.
- To reduce energy consumption through intelligent lighting.
- To simulate and verify the circuit operation using TinkerCAD.

3. Software Requirements

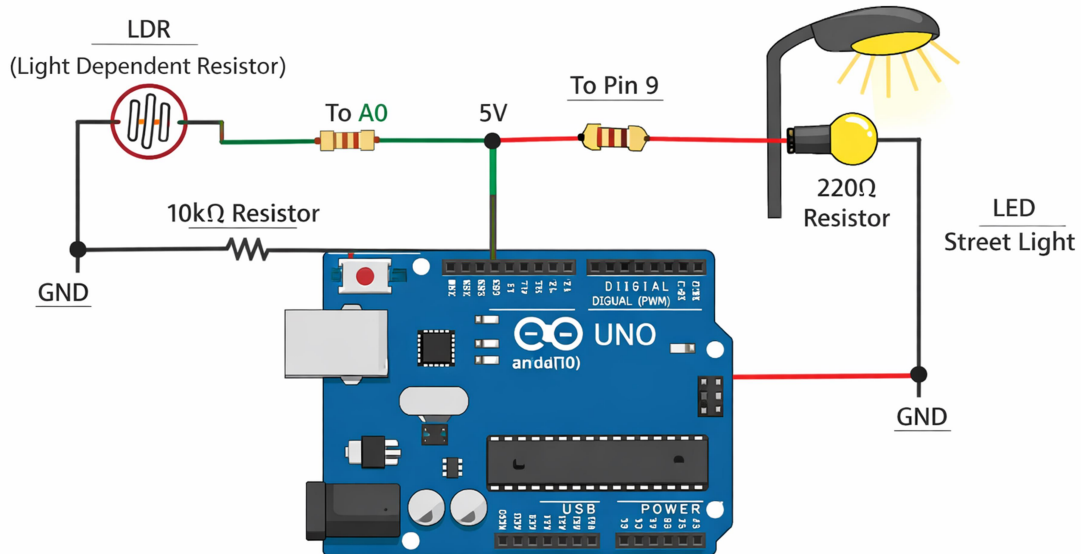
- **TinkerCAD Circuits (Online Simulation Tool).**
- Arduino IDE (built-in within TinkerCAD).
- Web browser such as Google Chrome or Microsoft Edge.
- Internet connection.

4. Steps to Access TinkerCAD

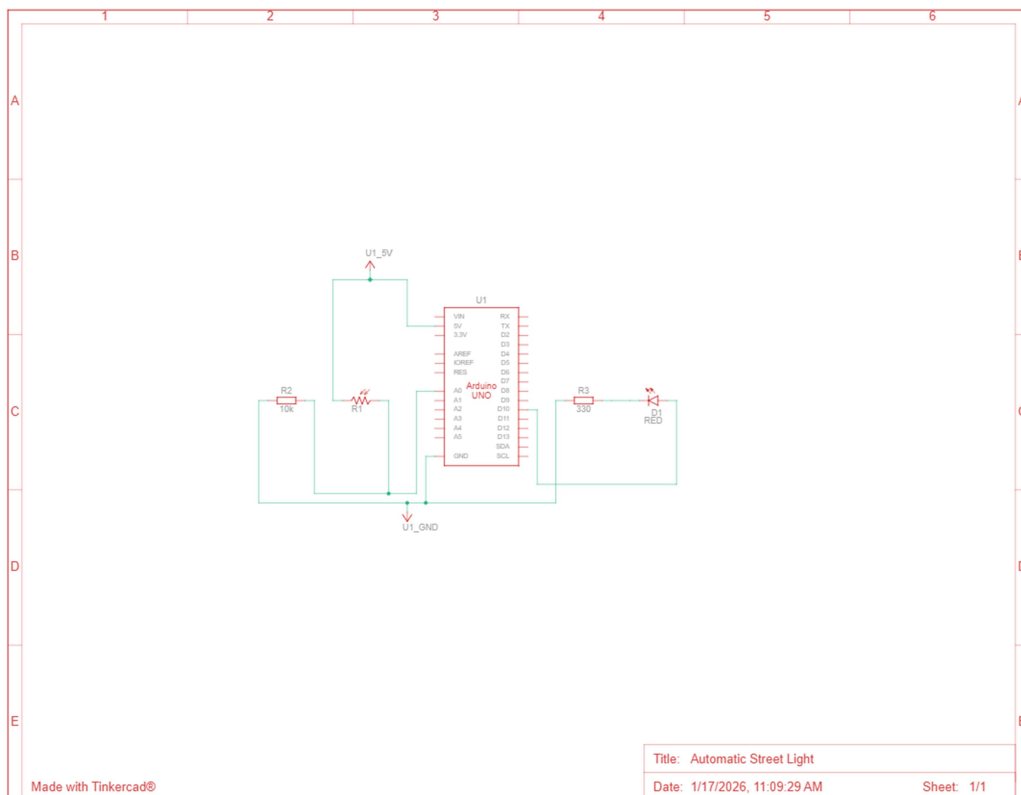
1. Open a web browser and go to the TinkerCAD website
2. Sign in using a Google account or create a new account
3. Click on **Circuits** from the TinkerCAD dashboard
4. Select **Create New Circuit**
5. Add components such as Arduino UNO R3, LDR, resistor, LED, and connecting wires
6. Write the Arduino code in the code editor
7. Click **Start Simulation** to test the circuit

Circuit Diagram

Automatic Street Light Using Arduino UNO

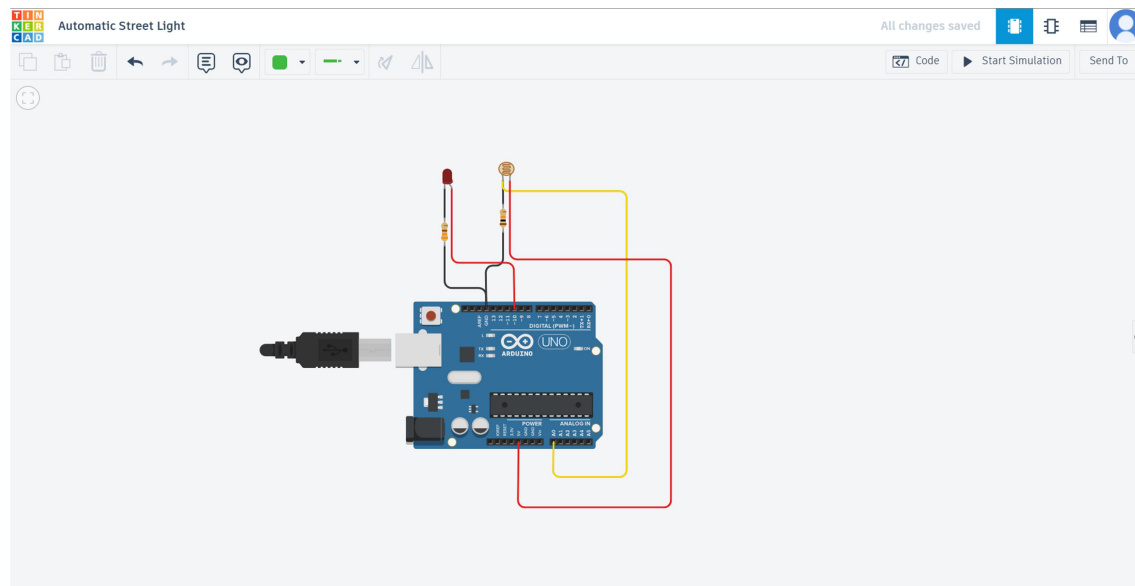


Schematic Diagram



Activity – Flipped Classroom

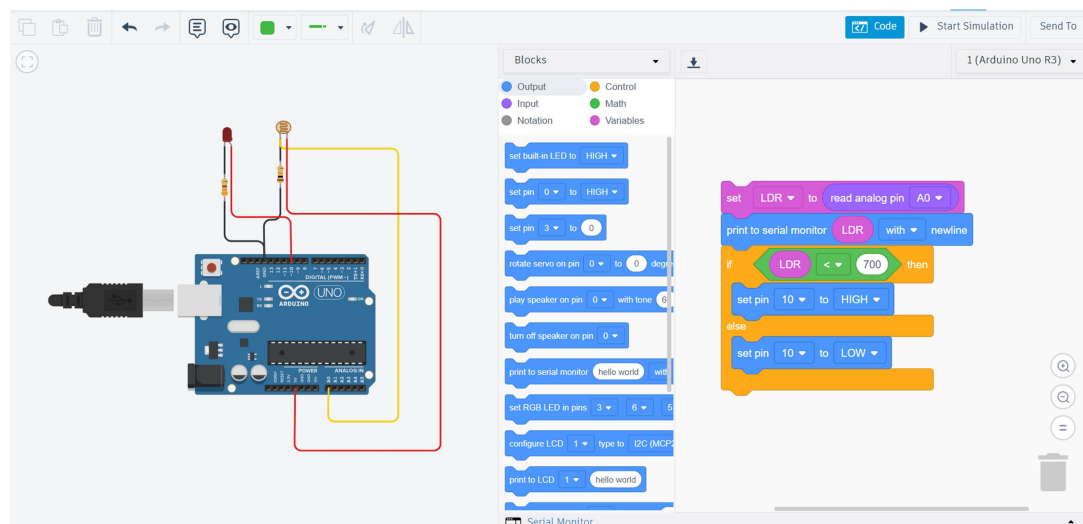
Circuit View



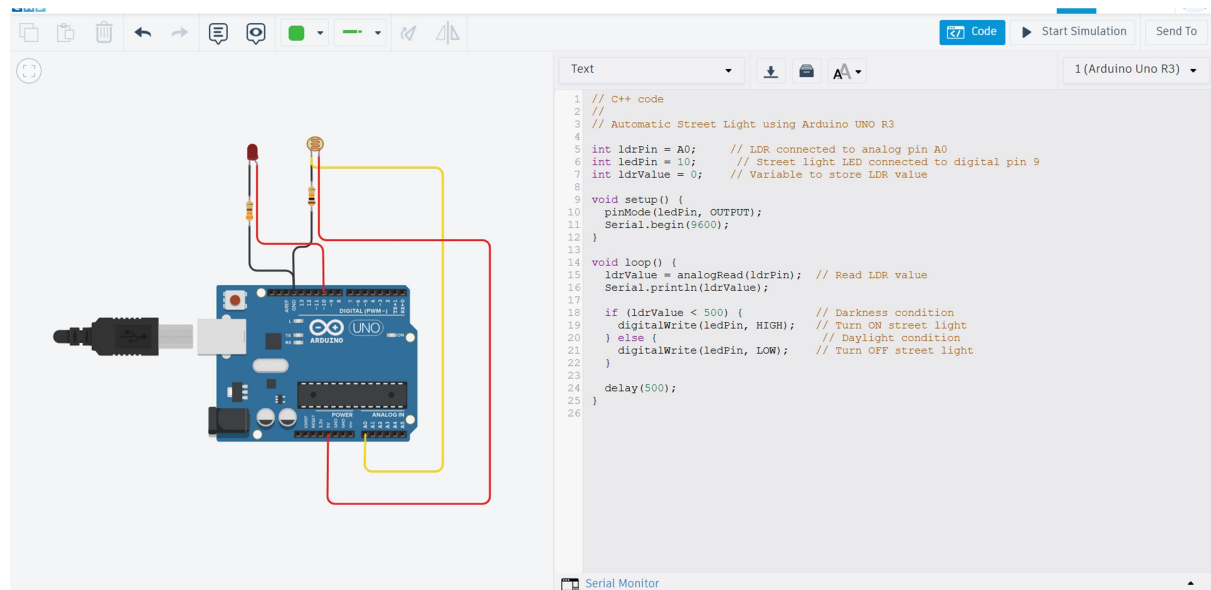
Components Required

Component List			Download CSV
Name	Quantity	Component	
U1	1	Arduino Uno R3	
R1	1	Photoresistor	
D1	1	Red LED	
R2	1	10 kΩ Resistor	
R3	1	330 Ω Resistor	

Code written in Blocks



Code Written in Text



5. Observation

- During bright light conditions, the LDR resistance decreases and the Arduino detects a higher analog value
- The street light (LED) remains OFF in daylight
- In low light or darkness, the LDR resistance increases
- The Arduino detects a lower analog value and switches the LED ON
- Smooth and automatic switching is observed during changes in light intensity

6. Uses of Automatic Street Light

- ✓ **Energy Conservation**
Automatic street lights turn ON only when required and remain OFF during daylight, reducing unnecessary power consumption and saving electricity.
- ✓ **Cost Reduction**
By minimizing energy usage and manual operation, automatic street light systems reduce electricity bills and maintenance costs.
- ✓ **Smart City Development**
These systems are widely used in smart city projects to improve infrastructure efficiency using automated and sensor-based technology.
- ✓ **Reduced Human Effort**
Manual switching of street lights is eliminated, reducing the need for human intervention and errors.
- ✓ **Improved Road Safety**
Street lights automatically illuminate during darkness, fog, or low-light conditions, improving visibility for pedestrians and drivers.

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- ✓ **Environment Protection**
Lower power consumption helps reduce carbon emissions and supports eco-friendly development.
- ✓ **Rural and Remote Area Lighting**
Automatic street lights are useful in rural areas where manual control is difficult or unreliable.
- ✓ **Highway and Road Lighting**
They are used on highways to ensure consistent lighting during night-time and poor weather conditions.
- ✓ **Industrial and Campus Lighting**
Industries, factories, educational institutions, and large campuses use

7. Result and Analysis

- The automatic street light system was successfully designed and simulated using **Arduino UNO R3 and TinkerCAD**, demonstrating accurate and reliable operation.
- The **LDR sensor effectively detected ambient light intensity**, producing different analog values for day and night conditions.
- During bright light conditions, the Arduino received higher analog values from the LDR and **kept the street light (LED) in OFF state**, confirming correct daylight detection.
- In low-light or dark conditions, the LDR resistance increased, resulting in lower analog values, and the **Arduino automatically switched ON the street light**.
- The threshold value programmed in the Arduino code provided **stable and consistent switching**, preventing frequent flickering of the LED.
- The system responded **quickly to changes in light intensity**, ensuring real-time control of the street light.
- Serial Monitor readings helped in **analyzing LDR values**, making it easier to calibrate the threshold level for different lighting conditions.
- The simulation verified that the system **reduces unnecessary power consumption**, as the light operates only when required.
- The circuit showed **no manual intervention**, confirming complete automation of the street lighting process.
- The project demonstrates a **practical and cost-effective solution** for street lighting, suitable for smart city and energy-saving applications.

8. Conclusion

The **Automatic Street Light using Arduino UNO R3** project was successfully designed, implemented, and simulated using the TinkerCAD virtual designing tool. The system effectively detects ambient light conditions using an LDR sensor and automatically controls the street light without the need for manual intervention. This confirms the reliability and accuracy of Arduino-based automation systems in real-time applications. The project clearly demonstrates how sensor data can be processed by a microcontroller to achieve intelligent decision-making.

The results obtained from the simulation show significant potential for energy conservation, as the street light remains OFF during daylight and turns ON only during low-light or night conditions. This not only reduces power consumption but also minimizes operational and maintenance costs. The simplicity of the circuit and code makes the system easy to implement and scalable for real-world applications such as highways, residential areas, and smart city infrastructure.

Furthermore, the use of TinkerCAD provided a safe and efficient environment for testing the system before physical deployment. Through this project, valuable knowledge was gained on Arduino programming, sensor interfacing, and automation concepts. Overall, the automatic street light system serves as a practical, eco-friendly, and cost-effective solution for modern lighting systems and highlights the importance of embedded systems in sustainable technological development.